Coco (Frontier System Project)

A digital everyday-assistance service for older adults, staffed by AI agents

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1. Problem & Value Hypothesis

My grandparents live on their own and prefer not to move in with their children. But living independently brings many small barriers: they have been targeted by scam messages, got lost in an unfamiliar area without knowing how to use a navigation app, cannot turn on their smart TV without help, and sometimes forget a pot on the stove. Each time, they have to call their family, and often no one is available. Many older adults face the same barriers, from finding information and reading emails to remembering appointments and medication.

Germany and China face the same trend. In **Germany**, **31%** of the population is aged 60+ (end of 2025) and **10%** of people aged 65–74 have never used the internet (2025). There are **33** people of pension age per 100 of working age today, possibly **61** by 2070, and by 2049 Germany is expected to lack **280,000 to 690,000** care workers. **China** already has **323.38 million** people aged 60+ (about **23%**, end of 2025), expected to exceed **400 million** by 2035. Only **53.7%** of them use the internet (December 2025), and about **60%** live alone or only with a spouse (2021). In both countries, families and carers cannot cover the growing need for everyday support.

Value hypothesis: a team of AI agents that remembers each person's needs, routines and family context can take over routine digital and daily tasks. Older adults stay independent, safe from scams and less lonely, while scarce human time goes to the care only humans can give. This is AI for Good in accessibility and healthy ageing.

2. The Startup: Service & Agent Team

[Coco] is a **service company**: it provides digital everyday assistance (not nursing care) through AI agents acting as employees, supervised by me as founder and care coordinator. Users get a **mobile app** (cross-platform, developed and tested on iPhone first), set up once by the family and opened with one tap (home-screen widget or AssistiveTouch shortcut). It starts in **English**(because the developer doesn't speak German) with text chat(easier to test); a **voice version** follows in January, and a Chinese version later. Reminders arrive as on-device notifications, also on an Apple Watch.

- **Front office (talks to the older person), built one by one in priority order:** (1) **Companion** with long-term memory (warm conversation based on remembered family and hobbies), (2) **Reminder assistant** (medication, appointments, short-term reminders), (3) **Scam guard**, (4) **Info helper** (searches the web and explains answers in plain language, with sources), (5) **Video helper** (plays YouTube and saved TikTok links inside the app via official embedded players), (6) **voice version**, then, if time allows, (7) **Navigator** ("Take me home" opens Google Maps with the route ready) and (8) **Mail helper** (Gmail summaries) and **Family liaison** (weekly summaries to relatives, with consent). Priorities 1–3 form the core; each next feature starts only once the previous ones pass their tests.
- **Back office (runs the company), built with coding tools such as Gstack:** from November, **engineer**, **code-review** and **QA** agents build the system and run the evaluation (Section 4); from December, a **scam analyst** turns anonymised reports into scam-trend warnings; **research/marketing** and **support** agents join with the business phase.

Boundaries: users always know they talk to an AI; emergencies (e.g., "I fell") are passed to family or emergency services (112); the agent never enters passwords or makes payments; if it cannot solve a task, it offers to call a family member, who can help via FaceTime screen sharing. **Later extensions:** smart-home control (e.g., TV, lights) via Home Assistant, Apple Watch health data (without medical interpretation), WhatsApp, an Android GUI agent, shopping and explaining official letters.

3. Agent Architecture & Workflow

- **LLM (Brain):** a frontier model via API (selected after early tests) understands requests, reasons and decides which tool to call or when to ask a clarifying question. Front-office roles share one assistant and one memory.

- **Memory (four types):** a **personal profile** (family, addresses, preferences), **recent events** with timestamps (e.g., “went to the supermarket yesterday”), summarized over time, **health information** (conditions, medications, doctor’s instructions), kept structured, separately protected and used only for reminders, never for advice, and **short-term reminders** (e.g., “the pot on the stove”), handled by a scheduler that repeats until the user confirms. Facts are confirmed before saving and tagged with source and date; retrieval ranks memories by relevance, recency and importance. Profile and events are first prototyped with the open-source system **GBrain**, with a custom module as fallback.
- **Tools (MVP):** web search; YouTube Data API plus saved video links, played through embedded players; Google Maps deep links and route data; read-only Gmail API; an on-device reminder scheduler; and a scam check combining LLM red-flag analysis, link checks against public threat databases and memory (e.g., saved family numbers). Tools are plug-ins, later replaceable by Chinese services (e.g., Amap, Douyin, WeChat); facts such as dates and routes come from tools, not the LLM.
- **Planning:** breaks goals into steps, e.g., “Prepare me for my doctor visit tomorrow” → check the appointment → list medications to bring → set a reminder to leave on time.
- **Workflow:** understand goal → retrieve memories → plan → call tools → check results → confirm sensitive actions → respond → update memory.
- **Skills:** each agent role, in the service and in the back office, is defined by reusable skills: instruction files with the steps, rules and output format for a task. This keeps behaviour consistent and testable.
- **Privacy by design:** conversations stay private and are never shown to family or company staff. Health information and reminder status are shared with family only if the older person agrees; emergencies are always passed on, as agreed at setup. The company sees only anonymised statistics (e.g., scam trends). Data is stored separately, encrypted and kept to a minimum (GDPR).

4. Success Metrics & Milestones

Product quality is measured in a simulated environment adapted from agent benchmarks such as CAR-bench, which I used in a previous agentic AI course. Three LLMs act as the **assistant**, a **simulated older user** (personas such as forgetful or vague) and an **evaluator**, supported by rule-based checks. Task types: **base tasks** (single-session requests), **long-term memory tasks** (one set per memory type, tested several sessions later, including updated facts) and **hallucination tasks** (missing information, failed tools, medical-advice requests). The scam check uses a labelled set of real English scam and harmless messages.

Metrics, compared with a **no-memory baseline**: task success rate, memory recall accuracy, false-memory and staleness rates, reminder timeliness, scam detection and false-alarm rates, hallucination rate and cost per task. **Agent-team performance:** hours saved and corrections needed.

Technical milestones (test tasks are re-run after each phase):

- **M1 (by 15 Nov):** basic question-answering agent; ~30 test tasks and user personas designed; first run as the no-memory baseline.
- **M2 (by 15 Dec):** GBrain integrated and tested on the memory tasks (target ≥80% recall); if it does not fit, a custom memory module is built.
- **M3 (December):** reminders and scam check (core), then search & explain and the video helper, with weekly evaluation runs.
- **M4 (January):** basic voice version (speech-to-text and text-to-speech around the same agent core); Maps and Gmail if time allows; final evaluation against the baseline (target ≥80% success on base tasks and a clear gain on memory tasks).

Business milestones (interviews, market choice, landing page, pilot users): to be decided after initial feedback.

5. Business Model

Users are older adults; **paying customers** are their adult children, who gain peace of mind through scam alerts and reminder summaries. Unlike Alexa or ChatGPT, [Coco] combines personal memory, plain-language help and a family loop. A family plan of about **€???/month** (to be tested) must stay clearly above the cost per user (LLM, memory hosting, APIs); because the staff are agents, personnel costs stay low.The app itself contains no advertising and never sells user data: older users are a frequent target of misleading ads, and a subscription-only model keeps the service trustworthy and simple to use. Later revenue: **anonymised scam-trend reports** for banks and telecom providers, and licensing to senior-care providers and assisted-living residences. [Coco] targets ageing societies such as Germany and China: the English prototype is later adapted for China; the first market will be confirmed after feedback.

Data sources: Destatis; National Bureau of Statistics of China; National Health Commission; CNNIC (57th report); Caixin (2024).